

Introduction to Contract Theory and Auctions

Hidden Information: Extensions & Applications

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Mechanisms

Let $\mathcal{M}$ be an arbitrary message space. A mechanism $\mu = (\mathcal{M}, g)$ consists of the message space $\mathcal{M}$ and a mapping $g : \mathcal{M} \rightarrow \mathcal{A}$, where for each $m \in \mathcal{M}$, $g(m) = (q(m), t(m))$.

In words, conditional on receiving a message m from the agent, the principal requests a production level $q(m)$ and a payment $t(m)$.

- When facing the mechanism μ , an agent of type θ chooses a best message $m^*(\theta)$ that is implicitly defined as:

$$t(m^*(\theta)) - \theta q(m^*(\theta)) \geq t(m) - \theta q(m) \quad \forall m \in \mathcal{M}$$

- The mechanism μ induces therefore an **allocation rule** $a : \Theta \rightarrow \mathcal{A}$ as follows:

$$a(\theta) = g(m^*(\theta)) = (q(m^*(\theta)), t(m^*(\theta)))$$

Direct Revelation Mechanisms

There is another class of mechanisms that is much more simple.

Direct Revelation Mechanisms

Let $\mathcal{M} = \Theta$ be the message space. A direct revelation mechanism $\mu = (\Theta, g)$ consists of the message space Θ and a mapping $g: \Theta \rightarrow \mathcal{A}$, where for each $\theta \in \Theta$, $g(\theta) = (q(\theta), t(\theta))$.

In words, the principal commits to offer the transfer $t(\theta)$ and the production level $q(\theta)$ if the agent announces the value θ for any $\theta \in \Theta$.

Truthful Revelation

A direct revelation mechanism μ is truthful if it is incentive compatible for the agent to announce his true type for any type, i.e. if we have the following set of restrictions:

$$t(\theta) - \theta q(\theta) \geq t(\theta') - \theta q(\theta'), \quad \forall \theta \in \Theta.$$

- Note that if we put $t(\theta_i) = t_i$ and $q(\theta_i) = q_i$ where $i = \ell, h$ we get the incentive constraints as previously defined.

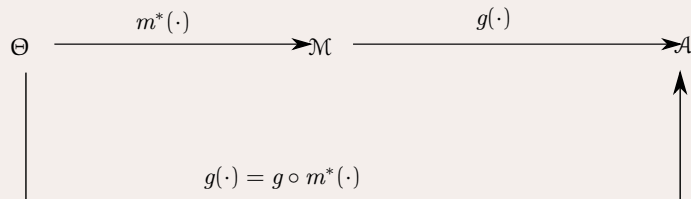
The revelation principle

The Revelation Principle

Any allocation rule $a(\theta)$ obtained with the mechanism μ can also be implemented with a truthful direct revelation mechanism μ

Proof of the Revelation Principle

Proof.



The indirect mechanism μ induces an allocation rule $a(\theta) = (q(m^*(\theta)), t(m^*(\theta)))$ from Θ into $\mathcal{A}$. By composition of g and m^* we can construct a direct revelation mechanism mapping Θ into $\mathcal{A}$, namely $g = g \circ m^*$, i.e. $g(\theta) = (q(\theta), t(\theta)) \equiv (q(m^*(\theta)), t(m^*(\theta)))$ for all $\theta \in \Theta$.

We now check that g is truthful. Note that if $m = m^*(\theta')$ then we have that

$$t(m^*(\theta)) - \theta q(m^*(\theta)) \geq t(m^*(\theta')) - \theta q(m^*(\theta')) \quad \forall (\theta, \theta') \in \Theta^2$$

Using the definition of g we get

$$t(\theta) - \theta q(\theta) \geq t(\theta') - \theta q(\theta'), \quad \forall (\theta, \theta') \in \Theta^2.$$

Applications: Credit rationing

- Financial markets are prone to adverse selection.
- Lender knows less about the risk-return of a project than the borrower.
- Lender is like a *used-car buyer*.
- Inefficiencies can arise in the form that *good quality projects* remain *unsold*, i.e. they are denied credit. → **credit rationing**.
- Setup:
 - ▶ Consider a population of risk-neutral borrowers.
 - ▶ Each borrower owns a project which requires an initial investment of $I = 1$.
 - ▶ Each project yields a random return $X \in \{R, 0\}$.
 - ▶ p denotes the probability that $X = R$.
 - ▶ Borrowers have no wealth, so they must obtain funds from an outside source.
 - ▶ A borrower can be of two types, $i = s, r$, s : safe, r : risky.
 - ▶ A borrower of type i has a project with return characteristics (p_i, R_i) .

Credit Rationing

- Assumptions

A1 $p_i R_i = m$ with $m > 1$.

A2 $p_s > p_r$ and $R_s < R_r$.

- Both Borrowers have same expected return, but different risk profile.
- A bank can offer to finance the initial investment in exchange for future repayment.
- Suppose there is only one bank, this bank has an **excess demand for funds**: amount available for funding is $\alpha < 1$. (Why we need this?)
- Unit mass of borrowers: proportion of safe borrowers is β and of risky borrowers is $1 - \beta$.
- Assume also that $\alpha > \max\{\beta, 1 - \beta\}$ so that available funds are sufficient to avoid crowding out either type of borrower.
- The first best contract (complete, symmetric information) is quite simple:
 - ▶ Bank is indifferent to whom to lend and lends the full amount α of funds
 - ▶ The contract specifies the repayment: D_i with $D_i = R_i$.
 - ▶ Whomever it lends to, repayment happens with probability p_i .

Credit Rationing: Simple contracts

- Suppose there is adverse selection.
- Initial literature focused on quite simple contracts: **single fixed-repayment** schemes.
- A single-fixed repayment scheme specifies an amount to be repaid D in exchange for the initial outlay $I = 1$.
- Assume that type- i borrowers apply for funds if and only if $D \leq R_i$.
- If $D > R_s$, only risky borrowers will consider applying.
 - ▶ In this case, the bank sets $D = R_r$ which gives a profit of $(1 - \beta)(p_r R_r - 1) = (1 - \beta)(m - 1)$.
- If $D \leq R_s$, both types of borrowers apply. Assuming each applicant has an equal chance of being financed (by a law of large numbers)
 - ▶ The bank finds it optimal to set $D = R_s$
 - ▶ Obtains a profit of $\alpha[\beta(m - 1) + (1 - \beta)(p_r R_s - 1)]$.

Credit Rationing arises under single fixed Repayment

- Note that $D = R_s$ is better than $D = R_r$ if and only if

$$\frac{\alpha\beta}{1-\beta} + \frac{\alpha(p_r R_s - 1)}{m-1} > 1$$

- If this is the case, then **credit rationing arises**: some risky borrowers cannot get credit.
- The risky borrowers are willing to accept a higher D , but the bank finds optimal not to raise D because it would lose the safe borrowers.
- Remark: This is an example of a **single-price** contract.
- Can the bank do better by offering more sophisticated contracts?

Type dependent fixed repayment contracts—Second Best

- A contract is defined by (x_i, D_i) where x_i is the probability of obtaining financing and D_i is the amount to be repaid.
- The bank solves the following problem (P)

$$\max_{(x_i, D_i)_{i=s,r}} \{ \beta x_s (p_s D_s - 1) + (1 - \beta) x_r (p_r D_r - 1) \} \quad (1)$$

subject to

$$0 \leq x_i \leq 1, \quad \forall i = s, r \quad (\text{Feasibility 1})$$

$$\beta x_s + (1 - \beta) x_r \leq \alpha \quad (\text{Feasibility 2})$$

$$x_i p_i (R_i - D_i) \geq 0 \quad \forall i = s, r \quad (\text{PC})$$

$$x_i p_i (R_i - D_i) \geq x_j p_i (R_i - D_j) \quad \forall i, j = s, r \quad (\text{IC})$$

- If the bank offers the first best menu $\{(x_s, D_s), (x_r, D_r)\} = \{(\alpha, R_s), (\alpha, R_r)\} \rightarrow$ Risky type mimics the safe type.

Solving the problem

- The previous consideration suggests the reduced problem (RP):

$$\max_{(x_i, D_i)_{i=s,r}} \{ \beta x_s (p_s D_s - 1) + (1 - \beta) x_r (p_r D_r - 1) \} \quad (2)$$

subject to

$$x_s p_s (R_s - D_s) \geq 0 \quad (\text{PC safe})$$

$$x_r p_r (R_r - D_r) \geq x_s p_r (R_r - D_s) \quad (\text{IC risky})$$

$$x_r \geq x_s \quad (\text{Monotonicity})$$

$$0 \leq x_i \leq 1 \quad i = s, r \quad (\text{Feasibility 1})$$

$$\beta x_s + (1 - \beta) x_r \leq \alpha \quad (\text{Feasibility 2})$$

- The reduced problem should be familiar:
 - ▶ PC constraint for the risky type does not bind.
 - ▶ IC constraint for the safe type does not bind.
 - ▶ Monotonicity of the allocation.
- The solution of (P) is equivalent to the solution of (RP).

Proof that P is equivalent to RP

- 1 Constraints of (P) imply those of (RP): From the two IC constraints:

$$x_r R_s - x_s R_s \leq x_r D_r - x_s D_s \leq x_r R_r - x_s R_r$$

which implies $x_r \geq x_s$.

- 2 At the solution to (RP), IC constraint holds as equality: otherwise the bank can increase D_r

► Here we use $x_r > 0$. If $x_r = 0$ then by step 1, $x_s = 0$, which cannot be a solution to (RP).

- 3 Solution of (RP) satisfies the constraints of (P)

► (PC safe) and (IC risky) constraints for (P) imply PC for the risky type:

$$x_r p_r(R_r - D_r) \geq x_s p_r(R_r - D_s) \geq x_s p_r(R_s - D_s) \geq 0$$

► Step 2 and monotonicity imply IC for safe type.

$$\begin{aligned} p_s(x_s(R_s - D_s) - x_r(R_s - D_r)) &= p_s(x_s R_s - x_r R_s + x_r D_r - x_s D_s) \\ &= p_s(x_r - x_s)(R_r - R_s) \\ &\geq 0 \end{aligned}$$

- Suppose that $x_s^* > 0$. From (PC safe), which must hold with equality, we get $D_s^* = R_s$.
- Substitute binding (IC risky) and (PC safe) into the objective function:

$$\max_{x_s, x_r} x_s (\beta(p_s R_s - 1) - (1 - \beta)p_r(R_r - R_s)) + (1 - \beta)x_r(p_r R_r - 1)$$

- Since we assumed $p_i R_i = m > 1$ at an optimal solution

$$x_r^* = 1 > \alpha \quad \text{why?}$$

- $x_s^* > 0$ implies

$$\beta(p_s R_s - 1) - (1 - \beta)p_r(R_r - R_s) \geq 0 \quad \text{why?}$$

- Together with monotonicity (Feasibility 2) we get

$$x_s^* = \frac{\alpha - (1 - \beta)}{\beta} < \alpha$$

- IC for risky type $(R_r - D_r^*) = x_s^*(R_r - D_s^*)$ and $D_s^* = R_s$ imply

$$D_r^* = x_s^* R_s + (1 - x_s^*) R_r > R_s = D_s^*$$

- If $\beta(p_s R_s - 1) - (1 - \beta)p_r(R_r - R_s) < 0$ then

$$x_s^* = 0$$

Solution: no credit rationing

- At this solution the bank trades off the rents extracted from risky borrowers with the ability to lend all its funds.
- When safe borrowers are numerous enough, bank wants to lend to them which leads to rent for risky borrowers.
- But rather than financing everybody by setting $D = R_s$ as in the single-price contract, the bank gives risky borrowers preferential access by setting $x_r^* > x_s^*$ in exchange for higher repayment.
- Credit rationing disappears: the only ones not fully funded are safe borrowers, but they are indifferent about being funded.
- Credit rationing reappears if we instead assume $p_r R_r < 1 < p_s R_s$.
 - ▶ Bank wants to turn away risky borrowers (i.e. set $x_r = 0$), but cannot do it without also denying the safe ones (since $x_r \geq x_s$).
 - ▶ Either everybody is funded (if $(\beta p_s + (1 - \beta)p_r) R_s \geq 1$) or nobody, i.e., there is financial collapse.

Non-linear pricing by a monopolist

- A monopolist is uninformed about the buyers' characteristics.
- What is the best pricing schedule according to quantity, quality, etc.?
- References: Mussa and Rosen (1978) & Maskin and Riley (1984).
- The model is similar to the delegation problem we studied a few lectures ago.
- You solved a version of this model for PS1.
- We will generalize quite a bit in several directions.

Two-type model

- Monopolist produces quantity (or quality) $q \geq 0$ at cost $c(q)$
- Cost function is C^2 (twice-continuously differentiable) with $c(0) = 0$, $c' \geq 0$, $c'' \geq 0$
- If monopolist sells a quantity q for t dollars, his payoff is: $w(q, t) = t - c(q)$
- Customer has two possible taste parameters $\theta_H > \theta_L > 0$.
 - ▶ Probability of θ_L is $p \in (0, 1)$.
- If Type θ_i buys quantity q and pays t , her payoff is $u(q, t, \theta_i) = v(q, \theta_i) - t$
- The valuation function v is twice C^2 with respect to q with

$$v(0, \theta_i) = 0, v_1 > 0, v_{11} \leq 0$$
$$v(q, \theta_H) > v(q, \theta_L) \quad \text{for any } q > 0$$

Notation

If $f: \mathbb{R}^n \rightarrow \mathbb{R}$ is a real-valued k -differentiable function we write

$$\frac{\partial f}{\partial x_i}(x) = f_i(x), \quad \frac{\partial^2 f}{\partial x_j \partial x_i}(x) = f_{ij}(x), \quad \frac{\partial^3 f}{\partial x_\ell \partial x_j \partial x_i}(x) = f_{ij\ell}(x)$$

and so on.

Surplus

- Total surplus available is $S(q, \theta_i) = v(q, \theta_i) - c(q)$
- Assume that trade is desirable $S_1(0, \theta_i) > 0$ for $i = L, H$.
- But there can be too much of a good thing, i.e. $\exists q$ such that $S(q, \theta_i) < 0 \forall q \geq q$
- Customer does not have to pay if she does not buy.
- The monopolist chooses how much to produce and possibly a non-linear price schedule.

First Best

- The monopolist solves the following problem for each $i = L, H$

$$\begin{aligned} \max_{q_i, t_i} \quad & t_i - c(q_i) \\ \text{s.t.} \quad & v(q_i, \theta_i) - t_i \geq 0. \end{aligned}$$

- The constraint must hold with equality, thus the problem becomes:

$$\max_{q_i \geq 0} S(q_i)$$

- The too much good assumption comes into play here: the previous problem is equivalent to

$$\max_{q_i} S(q_i) \quad \text{subject to } 0 \leq q_i \leq q$$

- Since the objective function is continuous and the constraint set is compact, then, by Weierstrass's Theorem, a solution to this problem exists.

Math detour

Theorem 1 (Weierstrass's Theorem)

Let $\mathcal{D} \subseteq \mathbb{R}^n$ be compact and $f: \mathcal{D} \rightarrow \mathbb{R}$ be a continuous function on $\mathcal{D}$. Then f attains a maximum and a minimum on $\mathcal{D}$

Theorem 2 (Heine-Borel)

A set $\mathcal{D} \subseteq \mathbb{R}^n$ is **compact** if and only if it is closed and bounded.

Definition 3

A set $\mathcal{D} \subseteq \mathbb{R}^n$ is **bounded** if there exists $k > 0$ such that $\|x\| \leq k$ for every $x \in \mathcal{D}$.

Here $\|x\|$ represents any norm in $\mathbb{R}^n$. Take for example $\|x\| = (\sum_{i=1}^n x_i^\rho)^{1/\rho}$ for $\rho \in [1, \infty)$. if $\rho = 2$ then we are talking about the usual **Euclidean norm**.

Definition 4

A set $\mathcal{D} \subseteq \mathbb{R}^n$ is **closed** if and only if for all sequences (x^k) such that $x^k \in \mathcal{D}$ for each k and $x^k \rightarrow x$ then $x \in \mathcal{D}$

Definition 5

Given an norm $\|\cdot\|$ on $\mathcal{D} \subseteq \mathbb{R}^n$ define the **distance induced by the norm** between $x, y \in \mathbb{R}^n$ as $d(x, y) = \|x - y\|$

Definition 6

A sequence x^k in $\mathcal{D} \subseteq \mathbb{R}^n$ is said to converge to a limit x (written $x^k \rightarrow x$) if for all $\epsilon > 0$ there exists an integer $K(\epsilon)$ such that for all $k \geq K(\epsilon)$ we have $d(x^k, x) < \epsilon$.

First-Best

- Let q_i^f be a solution
 - ▶ $q_i = 0$ cannot be a solution because trade at a positive quantity generates surplus, i.e. $S_1(0, \theta_i) > 0$.
 - ▶ Similarly, $q_i = q$ cannot be a solution since surplus is negative at q .
 - ▶ Thus, the solution must be interior:

$$S_1(q_i^f, \theta_i) = 0 \Leftrightarrow v_1(q_i^f, \theta_i) = c'(q_i^f), \quad i \in \{L, H\}$$

- if v is strictly concave on q ($v_{11} < 0$ or c is strictly convex $c'' > 0$) then the solution is unique.
- We can solve for the transfers from the constraint as

$$t_i^f = v(q_i^f, \theta_i) \quad i \in \{L, H\}$$

- Features:
 - ▶ Monopolists maximizes surplus and captures it all.
 - ▶ Efficient quantities are produced.
 - ▶ Price is set to capture entire consumer surplus.
 - ▶ This is **first-degree price discrimination**.

Single-Crossing

- Assume the **sorting condition or single-crossing**

$$v_1(q, \theta_H) > v_1(q, \theta_L)$$

- This is the Spence-Mirrlees condition previously discussed: The slope of the indifference curve for the high type is greater than the one for the low type.

- ▶ Note that an indifference curve is $u = v(q, \theta_i) - t$ for some u .
- ▶ Therefore, at any (q, t) , $\frac{dt}{dq} = v_1(q, t)$.
- ▶ An equivalent condition is **Increasing Differences**: If $q > q'$ and $\theta > \theta'$ then

$$v(q, \theta) - v(q', \theta) > v(q, \theta') - v(q', \theta')$$

- ★ Thus in this environment: Sorting condition $\Leftrightarrow$ Increasing differences.
- ★ Easy to prove: For $(\Rightarrow)$ use the Fundamental Theorem of Calculus; for $(\Leftarrow)$ use the definition of a derivative.
- In general the **Spence Mirrlees condition** is

$$\frac{\partial}{\partial \theta} \left(\frac{u_q}{u_t} \right) > 0 (< 0) \quad \text{for any } (t, q, \theta)$$

Implications of single-crossing

Claim

If single-crossing condition holds, then $q_H^f \geq q_L^f$ and $t_H^f \geq t_L^f$.

Proof.

Towards a contradiction, suppose that $q_H^f < q_L^f$. Then

$$c'(q_H^f) = v_1(q_H^f, \theta_H) > v_1(q_H, \theta_L) \geq v_1(q_L^f, \theta_L) = c'(q_L^f)$$

where the strict inequality is from single-crossing and the weak inequality is by concavity of v . The previous chain of inequalities implies that $c'(q_H^f) > c'(q_L^f)$ but since c is convex, this implies that $q_H^f \geq q_L^f$ which is a contradiction.

Furthermore, since v is increasing and by single-crossing: $t_H^f = v(q_H^f, \theta_H) \geq v(q_L^f, \theta_L) = t_L^f$. $\square$

Monopolistic Screening—Second Best Optimum

- As we know, under asymmetric information, if the monopolists offers the first best solution that we previously found, the high type will prefer to mimic the low type:

$$v(q_L^f, \theta_H) - t_L^f = v(q_L^f, \theta_H) - v(q_L^f, \theta_L) > 0$$

- Thus the monopolist would like to choose a price schedule $t(q)$ to maximize expected profits

$$\begin{aligned} \max_{t(q)} \quad & p(t(q_L) - c(q_L)) + (1 - p)(t(q_H) - c(q_H)) \\ \text{subject to} \quad & q_i \in \arg \max_{q \geq 0} \{v(q, \theta_i) - t(q)\} \quad \text{for } i = L, H \\ & v(q_i, \theta_i) - t(q_i) \geq 0 \quad \text{for } i = L, H \end{aligned}$$

- The problem is difficult as the choice set is a set of functions.
- But we can use the **revelation principle**.

Monopolistic Screening—Second Best

The revelation principle implies that the monopolist's problem can be formulated as

$$(P) \quad \max_{(q_L, t_L), (q_H, t_H)} p(t_L - c(q_L)) + (1 - p)(t_H - c(q_H)) \quad (3)$$

subject to

$$v(q_L, \theta_L) - t_L \geq v(q_H, \theta_L) - t_H \quad (4)$$

$$v(q_H, \theta_H) - t_H \geq v(q_L, \theta_H) - t_L \quad (5)$$

$$v(q_L, \theta_L) - t_L \geq 0 \quad (6)$$

$$v(q_H, \theta_H) - t_H \geq 0 \quad (7)$$

$$q_H \geq 0, q_L \geq 0 \quad (8)$$

- As we have seen, the problem is equivalent to a simpler one in which we drop all constraints except
 - ▶ IC constraint of type H .
 - ▶ PC constraint of type L .
 - ▶ Monotonicity constraint $q_H \geq q_L$.
 - ▶ Feasibility (positive quantities)

Relaxed problem

The relaxed problem is:

$$(RP) \quad \max_{(q_L, t_L), (q_H, t_H)} p(t_L - c(q_L)) + (1 - p)(t_H - c(q_H)) \quad (9)$$

subject to

$$v(q_H, \theta_H) - t_H \geq v(q_L, \theta_H) - t_L \quad (10)$$

$$v(q_L, \theta_L) - t_L \geq 0 \quad (11)$$

$$q_H \geq q_L \quad (12)$$

$$q_H \geq 0, q_L \geq 0 \quad (13)$$

Claim

Solution to the program (P) and (RP) are the same.

Proof

- 1 The constraint set of (P) is a subset of that of (RP). (4) and (5) imply that

$$v(q_H, \theta_L) - v(q_L, \theta_L) \leq t_H - t_L \leq v(q_H, \theta_H) - v(q_L, \theta_H)$$

Single crossing condition implies that $q_H \geq q_L$.

- 2 At the solution to (RP), (10) holds as an equality: Otherwise we could increase t_H a little bit.
- 3 Solutions to (RP) satisfy the constraint of (P): since (10) holds with equality and $q_H \geq q_L$, single crossing implies

$$t_H - t_L = v(q_H, \theta_H) - v(q_L, \theta_H) \geq v(q_H, \theta_L) - v(q_L, \theta_L)$$

which implies (4). Also (10), (11) and $\theta_H \geq \theta_L$ imply (6):

$$v(q_H, \theta_H) - t_H \geq v(q_L, \theta_H) - t_L \geq v(q_L, \theta_L) - t_L \geq 0$$

Proof (Cont.)

1 Let a^P solve (P) but not (RP)

- ▶ Step (1) implies a^P satisfies the constraints of (RP).
- ▶ At any solution a^R to (RP) the value of the objective function is strictly greater than its value at a^P .
- ▶ Then, a^R is not in the constraint set of (P), contradicting step (3).

2 Let a^R solve (RP) but not (P)

- ▶ Step (3) implies a^R is in the constraint set of (P)
- ▶ There is an a^P in the constraint set of (P) that gives strictly higher value for the objective function.
- ▶ Step (1) implies a^P is in the constraint set of (RP)
- ▶ Contradicts that a^R solves (RP)

Solving for the second best

- Notice that constraints (10) and (11) need to hold with equality in problem (RP).
- We will ignore (12) for the moment and check that the solution satisfies it.
- After substituting the binding constraints the problem becomes

$$\max_{q_L \geq 0, q_H \geq 0} \quad p(v(q_L, \theta_L) - c(q_L)) + (1 - p)(v(q_H, \theta_H) - c(q_H)) \\ - (1 - p)(v(q_L, \theta_H) - v(q_L, \theta_L))$$

- We have the usual setup: Second best maximizes expected efficiency minus expected information rent.

Second best solution

- For reasons that will become clear later, it is convenient to rewrite the problem as:

$$\max_{q_L \geq 0, q_H \geq 0} v(q_L, \theta_L) - c(q_L) + \frac{1-p}{p} (v(q_H, \theta_H) - c(q_H)) - \frac{1-p}{p} (v(q_L, \theta_H) - v(q_L, \theta_L))$$

- Assume the following term (called the virtual surplus) is strictly concave.

$$S^v(q, \theta_L, \theta_H) = (v(q, \theta_L) - c(q)) - \frac{1-p}{p} (v(q, \theta_H) - v(q, \theta_L))$$

- This implies the objective is concave, so the first order conditions (Kuhn-Tucker) will be necessary and sufficient.

Second Best Solution

Form the Lagrangean:

$$\begin{aligned} L(q_L, q_H, \lambda_L, \lambda_H) = & v(q_L, \theta_L) - c(q_L) - \frac{1-p}{p} (v(q_L, \theta_H) - v(q_L, \theta_L)) \\ & + \frac{1-p}{p} (v(q_H, \theta_H) - c(q_H)) + \lambda_L q_L + \lambda_H q_H \end{aligned}$$

A critical point must satisfy the following conditions:

$$\begin{aligned} \frac{1-p}{p} (v_1(q_H, \theta_H) - c'(q_H)) + \lambda_H &= 0 \\ (v_1(q_L, \theta_L) - c'(q_L) - \frac{1-p}{p} (v_1(q_L, \theta_H) - v_1(q_L, \theta_L)) + \lambda_L &= 0 \\ \lambda_L, \lambda_H, q_L, q_H &\geq 0, \quad \lambda_L q_L = \lambda_H q_H = 0 \end{aligned}$$

Second Best solution

- $q_H = 0 \Rightarrow v_1(0, \theta_H) - c'(0) \leq 0$, contradicting $S_1(0, \theta_H) > 0$.
- Therefore, $q_H > 0 \Rightarrow \lambda_H = 0$ and hence

$$v_1(q_H, \theta_H) - c'(q_H) = 0$$

- Therefore, $q_H > 0 \Rightarrow \lambda_L = 0$ and hence
- Similarly, if $q_L > 0$

$$v_1(q_L, \theta_L) - c'(q_L) - \frac{1-p}{p}(v_1(q_L, \theta_H) - v_1(q_L, \theta_L)) = 0$$

- and if $q_L = 0$, $\lambda_L \geq 0$ and

$$v_1(0, \theta_L) - c'(0) - \frac{1-p}{p}(v_1(0, \theta_H) - v_1(0, \theta_L)) \leq 0$$

Summary: second best

The unique global solution q_L^s, q_H^s is given by

$$v_1(q_H^s, \theta_H) = c'(q_H^s)$$

- If $S_1^v(0, \theta_L, \theta_H) \geq 0$

$$v_1(q_L^s, \theta_L) = c'(q_L) + \frac{1-p}{p}(v_1(q_L^s, \theta_H) - v_1(q_L^s, \theta_L))$$

- otherwise $q_L^s = 0$

The solution is also monotone

- We have to check if the solution satisfies constraint (12): $q_H^s \geq q_L^s$
- If $q_L^s = 0$ then it is satisfied trivially.
- So suppose $q_L^s > 0$. Then

$$v_1(q_L^s, \theta_L) = c'(q_L) + \frac{1-p}{p}(v_1(q_L^s, \theta_H) - v_1(q_L^s, \theta_L)) > c'(q_L^s)$$

- Remember $v_1(q_L^f, \theta_L) = c'(q_L^f)$. We claim that $q_L^s \leq q_L^f$. Towards a contradiction, $q_L^s > q_L^f$. Then, convexity of c implies that

$$v_1(q_L^s, \theta_L) > c'(q_L^s) \geq c'(q_L^f) = v_1(q_L^f, \theta_L)$$

- But concavity of v in q , i.e. $v_{11} \leq 0$, implies that $q_L^s \leq q_L^f$, a contradiction.
- Therefore we have $q_H^s = q_H^f \geq q_L^f \geq q_L^s$ and thus monotonicity is satisfied.

Features of the optimal second best.

- No distortion at the top
 - ▶ The high type consumes the efficient quantity but the low type's consumption is lower than the efficient level.
 - ▶ The high type obtains an information rent.
- High type consumes the same quantity as the first best but a lower price.
 - ▶ This gives the high type positive surplus, i.e. an information rent.
- The quantity and price for the low type are reduced.
 - ▶ This keeps the information rent obtained by the high type at a minimum.
- The distortion caused by the information rent received by the high type is decreasing in p .

More than two types

- Suppose there are $n \geq 2$ types such that:

$$\theta_n > \theta_{n-1} > \cdots > \theta_1$$

- Assume that $\Pr(\theta_i) = p_i > 0$, $\sum_{i=1}^n p_i = 1$.
- The rest of the model is the same:

$$v(q, \theta_{i+1}) \geq v(q, \theta_i), \quad v(0, \theta_i) = 0, \quad v_1 > 0, \quad v_{11} \leq 0, \quad v_1(q, \theta_{i+1}) > v_1(q, \theta_i)$$

- Results generalize
 - ▶ No distortion at the top.
 - ▶ Inefficiently low consumption for lower types.
 - ▶ No rent for the lowest type. Higher types receive information rent.
 - ▶ Downward incentives compatibility constraints bind.

General model: discrete types

The monopolist's problem is:

$$\max_{\{(q_i, t_i)\}_{i=1, n}} \sum_{i=1}^n p_i(t_i - c(q_i))$$

subject to

$$v(q_i, \theta_i) - t_i \geq v(q_j, \theta_i) - t_j, \quad \forall i, j \quad (14)$$

$$v(q_i, \theta_i) - t_i \geq 0, \quad \forall i \quad (15)$$

$$q_i \geq 0, \quad \forall i \quad (16)$$

Discrete types—Relaxed problem

The general problem is equivalent to the following relaxed problem:

$$\max_{\{(q_i, t_i)\}_{i=1, n}} \sum_{i=1}^n p_i(t_i - c(q_i))$$

subject to

$$v(q_i, \theta_i) - t_i \geq v(q_{i-1}, \theta_i) - t_{i-1}, \quad \forall i = 2, \dots, n \quad (17)$$

$$v(q_1, \theta_1) - t_1 \geq 0 \quad (18)$$

$$q_i \geq q_{i-1}, \quad i = 2, \dots, n \quad (19)$$

$$q_i \geq 0, \quad \forall i \quad (20)$$

Proof

First, we show that when $q_i \geq q_{i-1}$ holds, then (14) holds if and only if (17) and

$$v(q_i, \theta_i) - t_i \geq v(q_{i+1}, \theta_i) - t_{i+1}, \quad i = 1, \dots, n-1. \quad (21)$$

Clearly, (14) imply (17) and (21). To prove the other direction, fix an index $i \in \{2, \dots, n\}$ and take any $j < i$. Note that

$$\begin{aligned} v(q_i, \theta_i) - v(q_j, \theta_i) &= \sum_{k=j+1}^i v(q_k, \theta_i) - v(q_{k-1}, \theta_i) \\ &\geq \sum_{k=j+1}^i v(q_k, \theta_k) - v(q_{k-1}, \theta_k) \\ &\geq \sum_{k=j+1}^i t_k - t_{k-1} = t_i - t_j \end{aligned}$$

Similarly, for any $j > i$ (verify this!)

Proof

- 1 The Constraint set of (P) is a subset of that of (RP): (14) implies that

$$v(q_i, \theta_{i-1}) - v(q_{i-1}, \theta_{i-1}) \leq t_i - t_{i-1} \leq v(q_i, \theta_i) - v(q_{i-1}, \theta_i)$$

Single crossing condition implies that $q_i \geq q_{i-1}$.

- 2 At the solution to (RP), (17) holds as an equality for all i : Otherwise, we can increase all t_j , $j \geq i$, by some $\epsilon > 0$.
- 3 Solutions to (RP) satisfy the constraints of (P): Since (17) hold with equality and $q_i \geq q_{i-1}$, single crossing implies

$$t_i - t_{i-1} = v(q_i, \theta_i) - v(q_{i-1}, \theta_i) \geq v(q_i, \theta_{i-1}) - v(q_{i-1}, \theta_{i-1})$$

which implies (21). Our result from the previous slide implies (14). Showing that (14) and (18) imply (15) is left for you to verify.

Solving the n -types model

Clearly, constraints (17) and (18) are all binding. Therefore, the problem is

$$\max_{\{(q_i, t_i)\}_{i=1, n}} \sum_{i=1}^n p_i(t_i - c(q_i))$$

subject to

$$v(q_i, \theta_i) - t_i = v(q_{i-1}, \theta_i) - t_{i-1}, \quad \forall i = 2, \dots, n$$

$$v(q_1, \theta_1) - t_1 = 0$$

$$q_i \geq q_{i-1}, \quad i = 2, \dots, n$$

$$q_i \geq 0, \quad \forall i$$

As before, ignore for now the monotonicity constraint and form the Lagrangian

$$\begin{aligned}\mathcal{L} = & \sum_{i=1}^n p_i(t_i - c(q_i)) + \sum_{i=2}^n \lambda_i[v(q_i, \theta_i) - v(q_{i-1}, \theta_i) - t_i + t_{i-1}] \\ & + \lambda_1[v(q_1, \theta_1) - t_1] + \sum_{i=1}^n \mu_i q_i\end{aligned}$$

The Kuhn-Tucker conditions for optimality are:

$$\frac{\partial \mathcal{L}}{\partial q_i} = 0 \quad i = 1, \dots, n$$

$$\frac{\partial \mathcal{L}}{\partial t_i} = 0 \quad i = 1, \dots, n$$

$$\frac{\partial \mathcal{L}}{\partial \lambda_i} = 0 \quad i = 1, \dots, n$$

$$\mu_i q_i = 0, \mu_i \geq 0, q_i \geq 0 \quad i = 1, \dots, n$$

- We will focus on solutions such that $q_i > 0$ for all i .
- From the KT conditions, this implies that $\mu_i = 0$ for all i .

From the previous slide, the KT conditions imply that

- For $i < n$

$$\frac{\partial \mathcal{L}}{\partial q_i} = -p_i c'(q_i) + \lambda_i v_1(q_i, \theta_i) - \lambda_{i+1} v_1(q_i, \theta_{i+1}) = 0$$

$$\frac{\partial \mathcal{L}}{\partial t_i} = p_i - \lambda_i + \lambda_{i+1} = 0$$

- For $i = n$

$$\frac{\partial \mathcal{L}}{\partial q_n} = -p_n c'(q_n) + \lambda_n v_1(q_n, \theta_n) = 0$$

$$\frac{\partial \mathcal{L}}{\partial t_n} = p_n - \lambda_n = 0$$

- So, we have no distortion at the top:

$$v_1(q_n, \theta_n) = c'(q_n)$$

- Using $p_i - \lambda_i + \lambda_{i+1} = 0$ and $\lambda_n = p_n$ we get $\lambda_i = \sum_{k=i}^n p_k$
- Therefore, for $i < n$

$$v_1(q_i, \theta_i) - c'(q_i) = \frac{1 - P_i}{p_i} [v_1(q_i, \theta_{i+1}) - v_1(q_i, \theta_i)] > 0$$

where $P_i = \sum_{k=1}^i p_k$

- Note that the principal maximizes the sum of virtual surpluses:

$$S^v(q_i, \theta_i, \theta_{i+1}) = v(q_i, \theta_i) - c(q_i) - \frac{1 - P_i}{p_i} [v(q_i, \theta_{i+1}) - v(q_i, \theta_i)]$$

- This implies less than efficient q_i for all $i < n$.

- If the virtual surplus functions are strictly concave and
 - ▶ $(1 - P_i)/p_i$ (inverse hazard rate) decreasing in i
 - ★ Known as **monotone hazard rate property**
 - ▶ $v_1(q_{i+1}, \theta_{i+1}) - v_1(q_{i+1}, \theta_i) \geq v_1(q_{i+1}, \theta_{i+2}) - v_1(q_{i+1}, \theta_{i+1})$ then $q_{i+1} \geq q_i$.
 - ★ Proof: Towards a contradiction, suppose that $q_{i+1} \leq q_i$. Then we have that

$$\begin{aligned}
 0 &= v_1(q_i, \theta_i) - c'(q_i) - \frac{1 - P_i}{p_i} [v_1(q_i, \theta_{i+1}) - v_1(q_i, \theta_i)] \\
 &< v_1(q_{i+1}, \theta_i) - c'(q_{i+1}) - \frac{1 - P_i}{p_i} [v_1(q_{i+1}, \theta_{i+1}) - v_1(q_{i+1}, \theta_i)] \\
 &\leq v_1(q_{i+1}, \theta_{i+1}) - c'(q_{i+1}) - \frac{1 - P_i}{p_i} [v_1(q_{i+1}, \theta_{i+1}) - v_1(q_{i+1}, \theta_i)] \\
 &\leq v_1(q_{i+1}, \theta_{i+1}) - c'(q_{i+1}) - \frac{1 - P_{i+1}}{p_{i+1}} [v_1(q_{i+1}, \theta_{i+2}) - v_1(q_{i+1}, \theta_i)] = 0
 \end{aligned}$$

which is a contradiction.

- ▶ However, it is possible to have $q_i = 0$ for some $i < n$.
- ▶ In most papers, set of types is a continuum, e.g. $[\theta_L, \theta_H]$
 - ★ More technical but the results will be cleaner.